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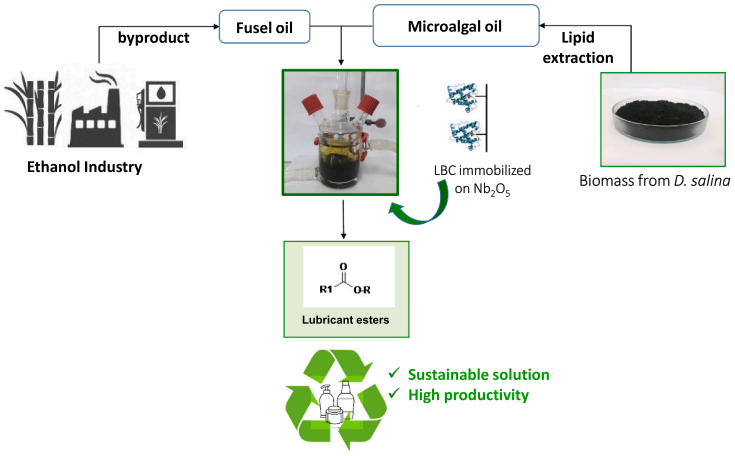
Full Length Article

Enzymatic catalysis: An environmentally friendly method to enhance the transesterification of microalgal oil with fusel oil for production of fatty acid esters with potential application as biolubricants

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GRAPHICAL ABSTRACT



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ABSTRACT

This work aimed at the enzymatic synthesis of fatty acid esters with potential application as biolubricants from microalgae oil and fusel oil using Burkholderia cepacia lipase immobilized on Nb_2O_5. The lipid-bearing Dunaliella salina biomass was grown in bubble column photobioreactors, yielding 100 g of biomass and 20 g of oil containing high content of saturated (37.8%) and monounsaturated fatty acids (41.2%) and a low content of polyunsaturated fatty acids (20.9%). The reaction assays were carried out in 50 mL glass reactor containing 10 g of medium composed by microbial oil and fusel oil components (ethyl, butyl, or isoamyl alcohols) and fusel oil as acyl acceptors at a fixed molar ratio of 1:8 (oil to alcohol) using iso-octane as solvent. Batch runs were performed at 45 °C under constant magnetic stirring (150 rpm) for a maximum of 120 h. Macaw palm oil was used as a control under the same reaction conditions. The biocatalyst was found to be efficient in converting the microalgal oil fatty acids into their corresponding alkyl esters attained yields ranging from 70.8 to 89.0% at 120 h reaction, depending on size of the alcohol chain. The kinematic viscosities of the esters produced confirmed the high transesterification yields, considering the viscosity of microalgae oil decreased from 60.1 to a range of 8.9–11.3 $m^2 s^{-1}$ (40 °C) after the enzymatic modification.

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1. Introduction

Lubricants are commonly composed of petroleum fractions (mineral oils) and additives. The imminent depletion of petroleum reserves, combined with the environmental impact potential and low biodegradability of mineral lubricants, necessitates the search for green alternatives [1]. This situation is further aggravated by the growing global demand for lubricants; production is projected to increase from 36.36 million tons in 2014 to 43.87 million tons in 2020, representing an annual growth rate of 2.4% [2].

Biolubricants are derived mainly from oil and fats and used as biodegradable alternatives to mineral lubricants to reduce environmental impacts in situations, where the probability of spills or leakage is high. Local and national environmental regulations and consumer demand for environmentally friendly materials have increased the awareness of the impacts of petroleum-derived products on the environment. As a result, the demand for biolubricants has continuously increased: these “green” products have sustained an annual market growth of 10% for the last 10 years. Biodegradability is broadly defined as the capability of a material to be broken down by microorganisms in up to one year. Completely degradable lubricants are able to essentially return to nature, whereas partially degradable lubricants contain one or more components that are not biodegradable [3].

The use of non-edible oils, that does not compete with the food industry, such as microbial oils to produce biolubricants is particularly advantageous [4]. Moreover, the costs of the vegetable oil are more expensive compared to mineral sources [5], and, generally these oils have high amounts of polyunsaturated fatty acids, which generates biolubricants with low oxidation stability [6]. A possible solution is to use an oil with higher amounts of the monounsaturated oleic acid (C18:1), and lower contents of linolenic acid (C18:3) [7], as for example, the microalgal oil used in this work obtained by cultivation of *Dunaliella salina*.

Biolubricant production from microbial oils has been regularly studied and reported on the literature. There are some important successful cases that indicate the high possibility to increase the marketing share of the biolubricants using microbial oils from yeasts such as *Cryptococcus curvatus* and *Rhodosporidium toruloides* [8–11].

The preference route reported to obtain biolubrificant is either by transesterification or hidro esterification using immobilized lipase as catalyst because of the numerous advantages of this catalyst over traditional homogeneous one [8–11]. Reactions cause little or no corrosion, product separation is simpler, and catalysts can be easily recovered and reused. In addition, problems related to purification steps are minimized and process costs are reduced, as fewer operations are needed to obtain the final product [3,12].

In this study, the catalyst chosen to perform the transesterification reaction of *D. salina* oil was the microbial lipase of *Burkholderia cepacia,* which was immobilized on Nb_2O_5. This biocatalyst exhibits high hydrolytic activity toward triglycerides regardless of the chain length of the fatty acids and does not have a particular positional specificity [13,14].

As acyl acceptor, fusel oil a byproduct of bioethanol industry was selected. Fusel oil is the least volatile fraction obtained in ethanol distillation and has an excellent potential for reusing due to its composition rich in higher alcohols and generated volume (120 million liters/year) [15,16]. Possible uses for this by-product include the production of purified higher alcohols by simple or fractional distillation and chemical or biotechnological synthesis of esters [17]. The major components of fusel oil are isoamyl alcohol (78.2%), butanol (10.6%), other secondary alcohols and ethanol (11.2%) [18]. Although fusel oil has been used to transesterify vegetable oils [18] no study has been described regarding the use of fusel oil as acyl acceptor for transesterification of microalga oil.

Considering the wide range of future possibilities for the biolubricants, this study aimed to evaluate the microbial oil from *Dunaliella salina* as an alternative oil feedstock for fatty acid esters production. Additionally, this work uses fusel oil components (ethanol, butanol, and isoamyl alcohol) and fusel oil as a new possibility of acyl acceptor for the microalgae lipids. The effect of molar ratio on the transesterification yield was initially evaluated using macaw palm oil, as reference oil, and the optimum value was subsequently used for the production of fatty acid esters with potential application as lubricant using the microbial oil.

2. Materials and methods

2.1. Materials

Marine microalga *Dunaliella salina* came from Seaweed Culture Collection (Oceanographic Institute - University of São Paulo) and was kindly provided by the Department of Biological Oceanography (São Paulo, Brazil). Lipase from *Burkholderia cepacia* (Lipase LBC) bought from Sigma-Aldrich (St. Louis, MO, USA) was immobilized into Nb_2O_5 according to Da Rós et al. [19] and used as biocatalyst in the synthesis of fatty acid esters with potential application as biolubricants [19]. Fusel oil was gently supplied by the distillery São Martinho S.A. (Iracema Unit, Iracemápolis, SP, Brazil) with the approximation of 68% wt. isoamyl alcohol, 8% wt. isobutyl alcohol, 0.4% wt. n-butyl alcohol, 0.5% wt. n-propyl alcohol, 9% wt. ethyl alcohol and 15% wt. water. Molecular sieves (3 Å) (sodium aluminum silicate) were purchased from Sigma-Aldrich. The high content of water initially present in the fusel oil has been reduced through molecular sieves of 0.32 Å (sodium aluminum silicate) type 13 X-BHD Chemicals. The concentration of water in the fusel oil was reduced from 15% to 4% as quantified by automatic Titler Karl Fisher (Koehler Model AKF5000). Macaw palm oil was supplied by the Association of Small Farmers D’Antas (Minas Gerais, Brazil), presenting the following fatty acid composition: 22.7 wt % palmitic acid, 2.8 wt% stearic acid, 53.4 wt% oleic acid, 17.7 wt% linoleic acid and 1.5% wt% linolenic acid with an average molecular weight of 864 g mol^{-1}. Other properties were: acid value (39.0 mg KOH g^{-1}), density (0.90 g/cm^3); viscosity (44.50 mm^2 s^{-1}) and water content (1.25%).

2.2. Cultivation of Dunaliella salina

Algal cultivations were carried out in bubble column photobioreactor (internal diameter of 150 mm, height of 1000 mm and 5.0 mm wall thickness) with 10 L working volume, equipped with LED light. The microalgae culture was inoculated at 20% (v/v) in Guillard f/2 culture media, supplemented with $NaNO_3$ (225 mg L^{-1}) and sodium acetate (4 g L^{-1}) according to Bredda et al. [20] during seven days at 25 ± 1 °C under 150 klux light intensity [20]. Microalgae cells were recovered by flocculation using 1 mol L^{-1} $Al_2(SO_4)_3$ solution. Microbial oil was extracted from biomass according to a modified Folch method, performed under ultrasound, with a mixture of chloroform: methanol (3:1 v/v). The mixture was sonicated for 10 min. This procedure was repeated two times [21]. Extracted lipids were dried in a rotatory evaporator to remove the residual solvent and subsequently dried at 60 °C until constant weight was attained.

2.3. Immobilization procedure

Hydrous niobium oxide (surface area of 42.15 m^2 g^{-1} and pore volume of 0.029 cm^3 g^{-1}) was prepared according to methodology previously described by Miranda et al. [22]. The support was submitted to a pre-treatment with nitric acid (1%) at 75 °C for 1 h to open the matrix pores and then silanized with γ-aminopropyltriethoxysilane (γ-APTS) and activated with glutaraldehyde [22]. Activated support was soaked into hexane under stirring (100 rpm) for 2 h at 25 °C, then excess of hexane was removed and lipase was added at a ratio of 1:4 g of enzyme per gram of support. PEG-1500 solution was added together

with the enzyme solution at a fixed amount (5 mg g^{-1} of support). Lipase-support system was maintained in contact for 18 h at 4 °C under static conditions. The immobilized lipase derivative was filtered, and hydrolytic activities of free and immobilized lipase were assayed by the olive oil emulsion method according to the modification described by Da Rós et al. [19]. To perform this work 15 g of the immobilized lipase was prepared having average hydrolytic activity of 1400 ± 12 U g^{-1}. One unit (U) of enzyme activity was defined as the amount of enzyme that liberates 1 μmol of free fatty acid per minute under the assay conditions.

2.4. Biocatalyst properties

Conformational changes, such as surface cavities in the support, set by the activating agent insertion were observed by scanning electron microscopy (SEM). Micrographs of support surfaces were obtained by using a Zeiss EVO MA 10 with EDS coupled LEO1450VP scanning electron microscope (Oxford Instruments Inca Penta FEET x3). Biochemical properties (optima pH and temperature), kinetic constants (K_m and V_{max}) and thermal stability were determined in the hydrolysis of emulsified olive oil. The effect of pH was evaluated at 40 °C in the range of 6.0–8.0 (buffer sodium phosphate 0.1 mol L^{-1}) while the effect of temperature was investigated in the range of 40–60 °C, using 0.1 mol L^{-1} buffer sodium phosphate at pH 6.5. The concentration of substrate (olive oil) was evaluated in the range of 20–70%, which corresponded to fatty acids concentrations of 744 to 2604 mmol L^{-1}. The apparent K_m and V_{max} values were calculated from Lineweaver–Burk plots, using the software Origin Pro 8. Thermal stability was measured by incubating immobilized derivative (0.05 g) in buffer phosphate pH 6.5 (0.1 mol L^{-1}) at 45 °C (temperature at which fatty acid esters production with potential application as biolubricants were conducted) for different time intervals. Samples were removed and assayed for residual activity as previously described, taking an unheated control to be 100% active. The rate of denaturation (k_d) and half-life time ($t_{1/2}$) were calculated by equations described by Da Rós et al. [19].

2.5. Fatty acid esters production with potential application as biolubricants

Initial tests were conducted with macaw palm oil in order to verify the influence of the oil/alcohol molar ratio (1:4, 1:6, and 1:8) on the transesterification yield. Considering previous research, the main components of the fusel oil (isoamyl alcohol, n-butyl alcohol and ethyl alcohol) were individually used as acyl acceptors [18]. The obtained results with the macaw palm oil indicated, for all tested acyl acceptors, that the excess of alcohol (1:8 oil/alcohol molar ratio) enhances the transesterification yields. Thus, such conditions were adopted to carry out the transesterifications reactions with microalgae oil.

Reactions were performed in a jacketed cylindrical glass reactor (5.5 cm height × 4 cm internal diameter, 70 mL capacity, coupled with a reflux condenser) containing 10 g of substrate (oil and acyl acceptor) at molar ratio of 1:8. Iso-octane was used as the reaction solvent at ratio of 40% to the reaction medium. Mixtures were incubated with immobilized lipase on Nb_2O_5 at a fixed proportion of 500 U per g of oil. Reactions were performed for a maximum period of 120 h, under constant stirring (150 rpm) at 45 °C. Aliquots at time intervals of 24 h were taken and subjected to analysis following the methodology described.

2.6. Analytical procedure

Unless stated differently, all experiments and analysis were performed at least three times. Data shown correspond to mean values. Fatty acid alkyl esters were purified according to methodology described by Santos et al. [23] using the adsorbent Amberlite BD10DRY (Rohm and Haas, Chemical Ltda) [23]. Samples of fatty acid esters and the Amberlite were transferred to a 50 mL Falcon tubes and placed in an ultrasound bath for 2 h. Then, the mixtures were centrifuged (1570 × g for 15 min) and residual solvent removed in a rotatory evaporator at 80 °C for 30 min. Alkyl esters contents were determined by gas chromatography using a Varian CG 3800 (Varian Inc. Corporate Headquarters, Palo Alto, CA, USA) equipped with flame ionization detector using a 5% DEGS CHR-WHP 80/100 mesh 6 ft 2.0 mm ID/800 OD stainless steel packaged column (Restek Frankel Commerce of Analytic Instruments Ltd., SP, Brazil), following methodology described by Ceron et al. [18]. The esters content were evaluated by proton nuclear magnetic resonance spectrometry (^{1}H NMR) in a Mercury 300 MHz-Varian spectrometer, with 5 mm glass tubes, using $CDCl_3$ as solvent and 0.3% tetramethylsilane (TMS) as internal standard. Residual glycerides (monoacylglycerol, MAG and diacylglycerol, DAG) were analyzed by an Agilent 1200 series HPLC (Agilent Technologies, USA) equipped with an evaporative light scattering detector and C18 Phenomenex (5 μm, 150 × 4.6 mm, 110 Å) column (Torrance, California). The viscosity was measured using a Brookfield LVDV-II viscometer (Brookfield Viscometers Ltd., England, UK) [23].

3. Results and discussion

3.1. Growth parameters and characterization of D. salina oil

Nine cultivations were carried out to obtain sufficient microalgal biomass for fatty acid esters production with potential application as biolubricants. A total of 103.50 g of biomass and 19.66 g of microalgal oil were produced using 90 L of growth medium. Table 1 shows the growth and production parameters of *D. salina*. (37.8%) and monounsaturated fatty acids (41.2%) and a low content of polyunsaturated fatty acids (20.9%).

D. salina oil had a high content of saturated (37.8%) and monounsaturated fatty acids (41.2%) and a low content of polyunsaturated fatty acids (20.9%) (Table 1). This fatty acid profile indicates good oxidative stability and the potential of *D. salina* oil for use in transesterification reactions. *D. salina* oil had similar fatty acid profile and physicochemical properties, such as viscosity (60.1 mm^2 s^{-1}), density (0.89 g mL^{-1}), and acid value (43.27 mg KOH g^{-1}), to macaw palm oil (see Section 2.1).

Table 1
Biomass and microalgal oil production by *Dunaliella salina* in a bubble column photobioreactor.

Parameter	Value
Biomass concentration (g L^{-1})	1.15 ± 0.40
Lipid concentration (mg L^{-1})	219 ± 0.30
Lipid yield (mg lipids mg^{-1} biomass)	0.19 ± 0.10
Volumetric lipid production rate (mg L^{-1} day^{-1})	31.28 ± 0.30
Volumetric biomass production rate (mg L^{-1} day^{-1})	164.28 ± 0.10
Lipid production rate (mg lipids mg^{-1} biomass day^{-1})	0.027 ± 0.08
Main fatty acids of the microalgal oil (wt%)	15.37% C16:0, 3.80% C16:1, 7.15% C18:0, 37.48% C18:1, 15.40% C18:2 and 2.31% C18:3

C16:0, palmitic acid; C16:1, palmitoleic acid; C18:0, stearic acid; C18:1, oleic acid; C18:2, linoleic acid; C18:3 linolenic acid.

3.2. Support and biocatalyst characteristics

Hydrous niobium oxide was used as a support matrix for immobilizing *B. cepacia* lipase by covalent bonding. Support activation was achieved by silanization of the hydroxyl groups using γ -APTS followed by transformation of the amino alkyl groups attached to the support surfaces into Schiff bases by treatment with glutaraldeyde [19].

The morphology and particle size distribution of silanized and activated Nb_2O_5 and immobilized derivative were determined by scanning electron microscopy (Fig. 1).

Silanized (Fig. 1a) and activated (Fig. 1b) Nb_2O_5 supports had similar specific surface areas and pore volumes. No differences in particle size and shape were observed. Fig. 1c shows that activated Nb_2O_5 particles were successfully covered by LBC, corroborating the high hydrolytic activity of the biocatalyst (1400 ± 12 U g^{-1}).

3.3. Biochemical properties and kinetic constants

The biochemical and kinetic properties of LBC immobilized on activated Nb_2O_5 are summarized in Table 2. Parameters were determined on the basis of data presented as Supplementary Material (Fig. S1 a–d).

The hydrolytic activity of the biocatalyst varied from 843.6 to 1166.2 U g^{-1} depending on the pH (Fig. S1a); the highest was obtained at pH 6.5. At pH 6.5, hydrolytic activity ranged from 1166.2 U g^{-1} at 35 °C to 1827.6 U g^{-1} at 55 °C (Fig. S1b). Da Rós et al. obtained similar values using the same biocatalyst: hydrolytic activity of 2033 U g^{-1} at pH 6.5 and 60 °C [19].

Biocatalyst activity was also measured as a function of fatty acid concentration (Fig. S1c). The reaction followed Michaelis–Menten kinetics, indicating that product formation did not inhibit enzyme activity. The regression showed that the experimental data fitted well the model of Michaelis-Menten, revealing the following K_m and V_{max} values, respectively, as 2796 mmol L^{-1} and 4411 μmol $g^{-1}min^{-1}$ with catalytic efficiency of 1.57. The catalytic efficiency (V_{max}/K_m) indicated the overall cumulative effect of V_{max} and K_m on the enzyme activity

Table 2
Biochemical and kinetics properties of *Burkholderia cepacia* lipase immobilized on Nb_2O_5.

Parameter	Value
Optimum pH	6.5
Optimum temperature °C	55
$t_{1/2}$ at 45 °C (h)	53.92 ± 0.49
K_m (mmol L^{-1})	2796 ± 97
V_{max} (μmol $g^{-1}min^{-1}$)	4411 ± 152

$t_{1/2}$, half-life; K_m, Michaelis–Menten constant; V_{max}, maximum reaction rate.

[19,24].

The thermal stability (inactivation constant (k_d) and the half-life time, $t_{1/2}$) of the biocatalyst was determined at 45 °C, the temperature at which ester synthesis was carried out (Fig. S1d). After 4 h of incubation, activity decreased by only 5%. The thermal deactivation constant (k_d) was 0.0185 h^{-1}, corresponding to a half-life time ($t_{1/2}$) of 53.92 h. These parameters confirm the positive influence of the immobilization method on the thermal stabilization of LBC, increasing the stability by 11 times in comparison to lipase in its free form. This $t_{1/2}$ value is similar to other derivatives of *B. cepacia* immobilized on different matrices, such as silica-carboxymethyl cellulose ($t_{1/2}$ = 70 h, 50 °C) as described by Martin et al. [24].

3.4. Transesterification kinetics and fatty acid esters composition

Fig. 2 shows the transesterification kinetics of microalgal oil and macaw palm oil with different acyl acceptors (ethanol, butanol, and isoamyl alcohol), as determined by gas chromatography. The fatty alkyl ester, monoacylglycerol (MAG), and diacylglycerol (DAG) contents obtained in the purified products from the conversion of microalgal and macaw palm oils into fatty alkyl esters are presented in Fig. 3 together with the fatty acids composition of each tested feedstock.

The results confirmed the potential of the biocatalyst to convert the

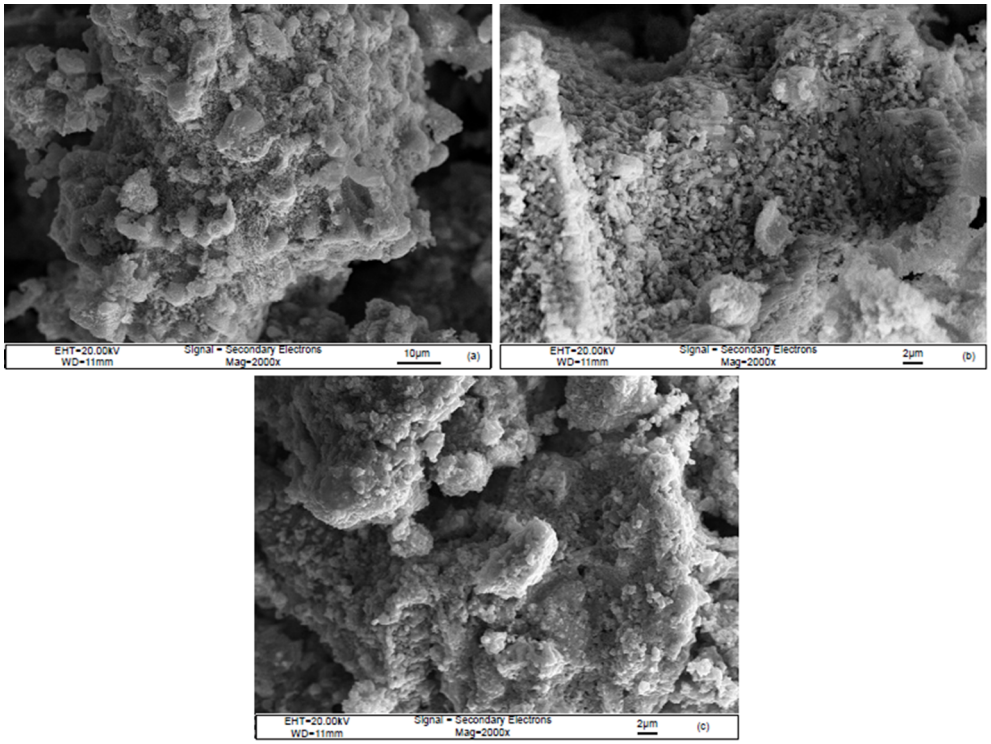


Fig. 1. Scanning electron micrographs (2000× magnification) of (a) silanized Nb_2O_5, (b) Nb_2O_5 activated with glutaraldehyde, (c) and *Burkholderia cepacia* lipase immobilized on activated Nb_2O_5.

fatty acids present in both feedstocks into alkyl esters independent of the acyl acceptor, although at different rates and yields. Overall, microalgal oil transesterification with ethanol, butanol, and isoamyl alcohol showed similar kinetics (Fig. 2a). Maximum ester formation was achieved in 96–120 h for all acyl acceptors and no marked differences were found on the final conversion values, which were in the range 75.5 to 82.1%. Much higher conversion (95.1–99.9%) were attained for macaw palm oil (Fig. 2b) at higher rate than the reactions performed with microbial oil as maximum ester formation were achieved between 72 and 96 h. Such difference can be attributed to the absence of any component in macaw palm oil that can interfere with the biocatalyst performance.

On the other hand, microbial oil particularly from microalga is recognized to have pigments, such as carotenoids or chlorophyll, which might reduce the transesterification yield [8]. Such interference has been already pointed out by some researchers, as for example in the study reported by Da Rós et al. [25] dealing with the transesterification of cyanobacterium *Trichormus* sp with ethanol catalyzed by a commercially available immobilized lipase (Novozym 435®). Although this immobilized lipase preparation is considered to be a robust biocatalyst, the presence of nonlipid compounds in the cynobacterium oil was found to affect the activity of lipase preparation. According to these authors [25], the adsorption of dyes or pigments onto the immobilizing support causes diffusion limitations and lipase inhibition, on account of the formation of a layer around the enzyme, which impose restriction of substrate access onto the enzyme active site, thus affecting its efficiency. This hypothesis was elucidated by submitting the original and recovered biocatalysts to a specific analysis (catalytic activity, FT-IR and SEM) to quantify changes that might have occurred in the biocatalyst after using it in the ethyl ester synthesis from the microbial oil [25]. Recently, the presence of chlorophyll in a particularly vegetable oil (Jupati oil) was also reported to interfere the biodiesel synthesis using a heterogeneous catalyst (sulfonated carbon-based catalyst), and such negative influence was overcome by using an adsorbent from Murumuru kernel shell, which was able to remove chlorophyll with 92.5% of efficiency [26]. In fact, there is a trend to study strategies to remove chlorophyll from microalga oils not only because it has serious negative impacts on the down-stream processing and quality of oil, but also to contribute on the consolidation of technology for the production of oil and other valuable products from microalgal biomass [27].

Regarding the formation of the alkyl esters in relation to the fatty acids present in each feedstock Fig. 3 shows that the profile of the formed alkyl esters corresponds to the fatty acids present in both feedstocks, and all fatty acids were converted at the same proportion without showing specificity, which can be explained by the non-specific character of the used biocatalyst. Specifically, the high formation of palmitate, oleate and linoleate esters corresponded to the fatty acids present at the highest proportions in both feedstocks, regardless the used acyl acceptor. The biocatalyst shows also poor specific to convert linolenic fatty acids present in both feedstocks at low proportions (< 2.5%).

Macaw palm oil afforded lower glyceride contents (1.55–4.05% MAG and 0.75–1.3% DAG) as a result of the high conversion. Microalgal oil, in contrast, produced of fatty acid esters with potential application as biolubricants with high MAG (10.5–21.2%) and DAG (2.8–6.2%) contents because the reaction did not achieve the stoichiometric equilibrium, resulting in lower alkyl ester formation.

Microbial oils have been successfully used to produce lubricant by enzymatic route. Papadaki et al. [8], for instance, produced wax esters from *Cryptococcus curvatus* oil and behenyl alcohol (a fatty alcohol), using lipase as biocatalyst. Papadaki et al [10] also obtained good results, 88% conversion, using *Cryptococcus curvatus* oil and neopentyl glycol, with *Candida rugosa* lipase as catalyst. Boviatsi et al. [11] reported good results of *Rhodosporidium toruloides* cultivated in sugar cane molasses. The work relates 85% yield in the esterification of the microbial oil with neopentyl glycol.

3.5. Transesterification of microalgal oil with simulated fusel oil and fusel oil

Having tested each acyl acceptor separately, the transesterification of microalgal oil with simulated and real fusel oil were also performed. Simulated fusel oil was composed of 78.2% isoamyl alcohol, 10.6% *n*-butanol, and 11.2% ethanol. Reactions were conducted for 120 h using *D. salina* oil at a 1:8 oil/alcohol molar ratio and LBC immobilized on Nb_2O_5 as biocatalyst.

Esters obtained from the transesterification of microalgal oil with fusel oil, and simulated fusel oil were qualitatively analyzed using nuclear magnetic resonance (NMR). Ethyl esters were identified by the presence of a quartet at 4–4.5 ppm (Fig. S2b, Supplementary Material), butyl and isoamyl esters by the presence of triplets (Fig. S2c and d, Supplementary Material). Superposition of low-intensity quartets and triplets characterize the presence of butyl and isoamyl esters in fusel oil alkyl esters (Fig. S2e and f, Supplementary Material).

Fusel oil and the simulated fusel oil esters were purified and analyzed by high-pressure liquid chromatography (HPLC). Residual MAG

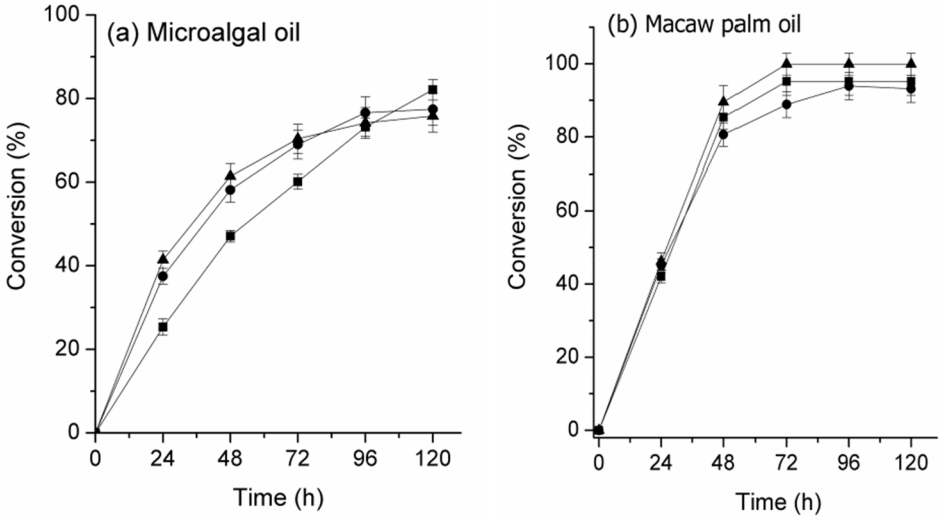


Fig. 2. Conversion of microalgal oil (a) and macaw palm oil (b) into ethyl (▲), butyl (■) and isoamyl (●) esters using *Burkholderia cepacia* lipase immobilized on Nb_2O_5 as catalyst (molar ratio oil to alcohol 1:8 at 45 °C).

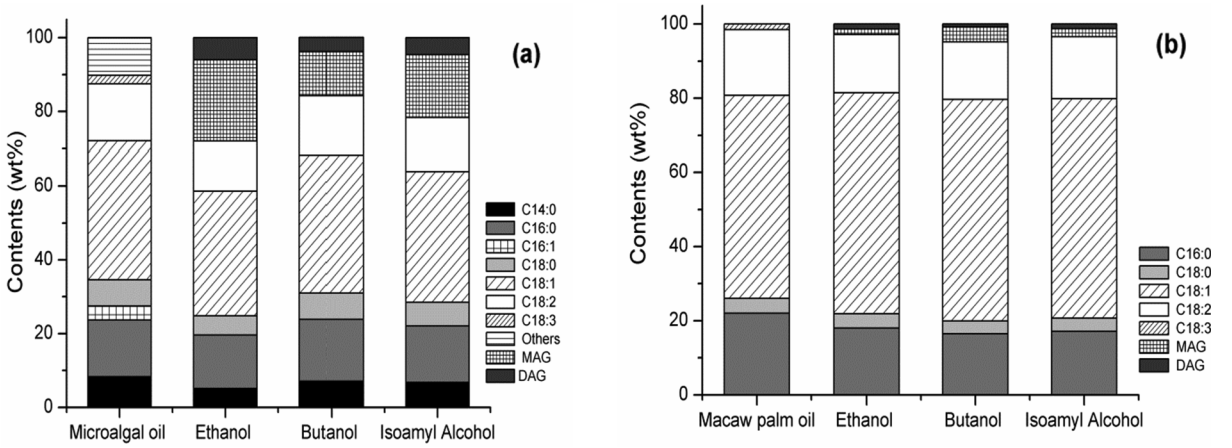


Fig. 3. Feedstocks fatty acids composition, fatty acid alkyl ester, monoacylglycerol (MAG), and diacylglycerol (DAG) contents obtained by transesterification of microalgal oil (a) and macaw palm oil (b) with ethanol, butanol, or isoamyl alcohol using *B. cepacia* lipase immobilized on Nb_2O_5 as biocatalyst.

Table 3
Residual monoacylglycerol (MAG) and diacylglycerol (DAG) contents and viscosity of purified products obtained by transesterification of microalgal oil with simulated or real fusel oil at a 1:8 oil/alcohol molar ratio using *Burkholderia cepacia* lipase immobilized on Nb_2O_5 as biocatalyst.

Alcohol	Ester contents (%)	MAG (wt%)	DAG (wt%)	Kinematic Viscosity ($mm^2\ s^{-1}$)
Simulated fusel oil	90.0 ± 0.9	7.8 ± 0.6	2.2 ± 0.2	8.6 ± 0.2
Fusel oil	89.0 ± 1.2	8.5 ± 0.7	2.6 ± 0.4	7.8 ± 0.5

and DAG contents and viscosity are presented in Table 3.

High ester contents (> 89.0%) and products with similar viscosities (7.8 and 8.6 $mm^2\ s^{-1}$) were obtained using real and simulated fusel oil. Results show that transesterification of microalgal oil with fusel oil using LBC immobilized on Nb_2O_5 as biocatalyst can afford favorable yields in comparison with data reported. The process is a promising alternative for fatty acid esters production with potential application as biolubricants from a renewable source.

4. Conclusion

The feasibility of producing alkyl esters with potential application as biolubricants by enzymatic transesterification of *Dunaliella salina* oil with higher alcohols (ethanol, butanol, or isoamyl alcohol) or alcohol mixtures (fusel oil) under batch runs was demonstrated. *D. salina* oil has a suitable fatty acids composition, which gives an appropriate balance between saturated and mono- and polyunsaturated fatty acids that helps to obtain fatty alkyl acids with good oxidative stability. *Burkholderia cepacia* lipase immobilized on Nb_2O_5 showed suitable catalytic properties and thermal stability, which allow obtaining conversion of the microalga oil into the corresponding alkyl esters with yields ranging from 70.8 to 89.0% at 120 h reaction, depending on size of the alcohol chain. This result also shows the potential of fusel oil to be used as acyl acceptor to obtain fatty esters from microalgae lipids in future applications to biolubricant production. To enhance the transesterification it is recommended to apply strategies for removing non-lipid compounds, such as chlorophyll.

CRediT authorship contribution statement

Ana Paula T. Da Silva: Data curation, conceptualization, Investigation, Writing - original draft, Writing - review & editing. **Eduardo H. Bredda:** Data curation, Methodology, Formal analysis, Software, Writing - original draft, Writing - review & editing. **Heizir F. de Castro:** Validation, Visualization, Supervision, Writing - original draft, Writing - review & editing. **Patrícia C. M. Da Rós:** Visualization, Resources, Supervision, Writing - original draft, Writing - review & editing, Project administration.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2020.117786.

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